1

Question Number : 50 Question Id : 69112150 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A tub is filled with water and a wooden cube 10 cm × 10 cm × 10 cm is placed in the water. The wooden cube is found to float on the water with a part of it submerged in water. When a metal coin is placed on the wooden cube, the submerged part is increased by 3.87 cm. The mass of the metal coin is _____ gram.

(Take water density as 1 g/cm^3 and density of wood as 0.4 g/cm^3)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id :	6911215
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	6911215
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 69112151 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The mass of iron converted into Fe_3O_4 by the action of 18 g of steam is :

(Given : Molar mass of H, O and Fe are 1, 16 and 56 g mol^{-1} respectively)

Assume iron is present in excess :

Options :

691121171. 2.1 g

691121172. 4.2 g

691121173. 21 g

691121174. 42 g

Question Number : 52 Question Id : 69112152 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

What is the energy (in J atom^{-1}) required for the following process ?

$Li^{2+}(g) \rightarrow Li^{3+}(g) + e^-$

(Take the ionization energy for the H atom in the ground state as 2.18×10^{-18} J atom^{-1})

Options :

691121175. 8.72×10^{-18}

691121176. 1.962×10^{-18}

691121177. 1.962×10^{-17}

691121178. 6.54×10^{-17}

Question Number : 53 Question Id : 69112153 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement (I) : The correct sequence of bond lengths in the following species is :

$O_2^+ < O_2 < O_2^- < O_2^{2-}$

Statement (II) : The correct sequence of number of unpaired electrons in the following species is :

$O_2 > O_2^+ > O_2^- > O_2^{2-}$

In the light of the above statements, choose the **correct answer** from the options given below :

Options :

691121179. Both **Statement I** and **Statement II** are true

691121180. Both **Statement I** and **Statement II** are false

691121181. **Statement I** is true but **Statement II** is false

691121182. **Statement I** is false but **Statement II** is true

Question Number : 54 Question Id : 69112154 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following data.

(i) $2Al(s) + 6HCl(aq) \rightarrow Al_2Cl_6(aq) + 3H_2(g) + 1200$ kJ/mol

(ii) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g) + 164$ kJ/mol.

(iii) $HCl(g) + aq \rightarrow HCl(aq) + 83$ kJ/mol.

(iv) $Al_2Cl_6(s) + aq \rightarrow Al_2Cl_6(aq) + 663$ kJ/mol

The enthalpy of formation of anhydrous solid Al_2Cl_6 is :

Options :

691121183. -648 kJ mol^{-1}

691121184. $-1350\ \text{kJ mol}^{-1}$

691121185. $-2002\ \text{kJ mol}^{-1}$

691121186. $-1527\ \text{kJ mol}^{-1}$

Question Number : 55 Question Id : 69112155 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

19.5 g of fluoro acetic acid (molar mass = 78 g mol^{-1}) is dissolved in 500 g of water at 298 K. The depression in the freezing point of water was 1°C. What is K_a of fluoro acetic acid ? (For water, $K_f = 1.86\ \text{K kg mol}^{-1}$). Assume molarity and molality to have same values.

Options :

691121187. 10^{-6}

691121188. 4×10^{-4}

691121189. 3×10^{-5}

691121190. 3×10^{-3}

Question Number : 56 Question Id : 69112156 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The solubility product constants of Ag_2CrO_4 and AgBr are $32x$ and $4y$ respectively at 298 K.

The value of $\left(\frac{\text{molarity of } Ag_2CrO_4}{\text{molarity of AgBr}}\right)$ can be expressed as :

Options :

691121191. $\frac{2\sqrt[3]{x}}{y}$

691121192. $2\sqrt{\frac{x}{y}}$

691121193. $\sqrt{\frac{x}{y}}$

691121194. $\frac{\sqrt[3]{x}}{\sqrt{y}}$

Question Number : 57 Question Id : 69112157 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An electrochemical cell is constructed using half cells in the direction of spontaneous change

$Fe(OH)_2(s) + 2e^- \rightarrow Fe(s) + 2OH^-(aq)$ $\quad E^\theta = -0.88$ V

and $AgBr(s) + e^- \rightarrow Ag(s) + Br^-(aq)$ $\quad E^\theta = +0.07$ V

Which of the following option is **correct** ?

Options :

691121195. Overall reaction $Fe(s)+2OH^-(aq)+2AgBr(s) \rightleftharpoons Fe(OH)_2(s)+2Ag(s)+2Br^-(aq)$

691121196. $E^\theta_{cell} = -0.95$ V

691121197. Fe is reduced in the electrochemical cell

691121198. E^θ_{cell} is an extensive property

Question Number : 58 Question Id : 69112158 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

$t_{100\%}$ is the time required for the 100% completion of the reaction while $t_{½}$ is the time required for 50% of the reaction to be completed. Which of the following option correctly represents the relation between $t_{100\%}$ and $t_{½}$ for zero and first order reactions respectively ?

Options :

691121199. $t_{100\%} = (t_{½})^2$ and $t_{100\%} = (t_{½})^{-\infty}$

691121200. $t_{100\%} = 2t_{½}$ and $t_{100\%} = (t_{½})^{\infty}$

691121201. $t_{100\%} = 2t_{½}$ and $t_{100\%} = (2t_{½})^2$

691121202. $t_{100\%} = (t_{½})^{\infty}$ and $t_{100\%} = 2t_{½}$

Question Number : 59 Question Id : 69112159 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement (I) : The first ionisation enthalpy of the elements Na, Mg, Cl and Ar follows the order Na > Mg > Cl > Ar.

Statement (II) : Among Ca, Al, Fe and B, the third ionisation enthalpy is very high for Ca.

In the light of the above statements, choose the **correct answer** from the options given below :

Options :

691121203. Both **Statement I** and **Statement II** are true

691121204. Both **Statement I** and **Statement II** are false

691121205. **Statement I** is true but **Statement II** is false

691121206. **Statement I** is false but **Statement II** is true

Question Number : 60 Question Id : 69112160 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement (I) : Oxidising power of halogens decreases in the order $F_2 > Cl_2 > Br_2 > I_2$, which is the basis of "Layer test".

Statement (II) : "Layer test" to identify Br_2 and I_2 in aqueous solution involves the oxidation of bromide or iodide into Br_2 or I_2 respectively with Cl_2, which is a type of displacement redox reaction.

In the light of the above statements, choose the **correct answer** from the options given below :

Options :

691121207. Both **Statement I** and **Statement II** are true

691121208. Both **Statement I** and **Statement II** are false

691121209. **Statement I** is true but **Statement II** is false

691121210. **Statement I** is false but **Statement II** is true

Question Number : 61 Question Id : 69112161 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following sets includes all the species that will change the orange colour of $K_2Cr_2O_7$ in acidic medium ?

Options :

691121211. $Fe^{2+}, Sn^{2+}, I^-, S^{2-}$

691121212. $S^{2-}, Fe^{3+}, I^-, C_2O_4^{2-}$

691121213. $Fe^{2+}, NO_2^-, SO_2, Sn^{4+}$

691121214. $Fe^{3+}, SO_4^{2-}, S^{2-}, Sn^{4+}$

Question Number : 62 Question Id : 69112162 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

	List - I Chromium (III) Complexes (en = ethylene diamine)		**List - II** $\Delta_0(cm^{-1})$
A.	$[Cr(CN)_6]^{3-}$	I.	15,060
B.	$[CrF_6]^{3-}$	II.	17,400
C.	$[Cr(H_2O)_6]^{3+}$	III.	22,300
D.	$[Cr(en)_3]^{3+}$	IV.	26,600

Choose the **correct** answer from the options given below :

Options :

691121215. A-I, B-II, C-III, D-IV

691121216. A-II, B-III, C-IV, D-I

691121217. A-III, B-IV, C-I, D-II

691121218. A-IV, B-I, C-II, D-III

Question Number : 63 Question Id : 69112163 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement (I) : 1,2,3-Trihydroxypropane can be separated from water by simple distillation.

Statement (II) : An azeotropic mixture cannot be separated by fractional distillation.

In the light of the above statements, choose the **correct answer** from the options given below :

Options :

691121219. Both **Statement I** and **Statement II** are true

691121220. Both **Statement I** and **Statement II** are false

691121221. **Statement I** is true but **Statement II** is false

691121222. **Statement I** is false but **Statement II** is true

Question Number : 64 Question Id : 69112164 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement (I) : Benzyl chloride reacts faster in S_N1 mechanism than ethyl chloride.

Statement (II) : Ethyl carbocation intermediate is less stabilized by hyperconjugation than benzyl carbocation by resonance.

In the light of the above statements, choose the **correct answer** from the options given below :

Options :

691121223. Both **Statement I** and **Statement II** are true

691121224. Both **Statement I** and **Statement II** are false

691121225. **Statement I** is true but **Statement II** is false

691121226. **Statement I** is false but **Statement II** is true

Question Number : 65 Question Id : 69112165 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In IUPAC nomenclature, the **correct** order of decreasing priority of functional group is :

Options :

691121227. $-CONH_2, >C=O, -CHO, -NH_2, -C\equiv C-$

691121228. $-CONH_2, -COOCH_3, -CHO, -NH_2, -OH$

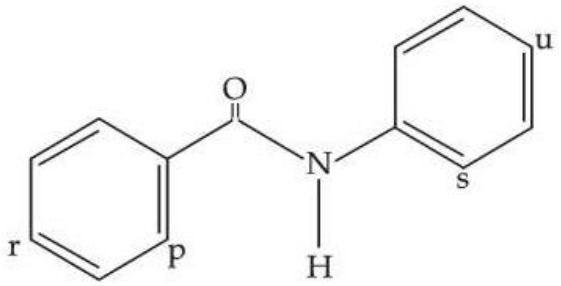
691121229. $-CONH_2, -CHO, >C=O, -NH_2, -C\equiv C-$

691121230. $-CONH_2, -CHO, -CN, -NH_2, -C\equiv C-$

Question Number : 66 Question Id : 69112166 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For the given molecule, "*x*", the preferred site for the attack of the electrophile is :

x =



Options :

691121231. Predominantly at "r"

691121232. "r" and "u"

691121233. "p" and "s"

691121234. Predominantly at "u"

Question Number : 67 Question Id : 69112167 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

	List - I		**List - II**
	Mixture of Compounds		**Reagent used to distinguish**
A.	Diethyl amine + Ethyl amine	I.	Bromine water
B.	Acetaldehyde + Acetone	II.	$CHCl_3$ + KOH, Δ
C.	Ethanol + Phenol	III.	Neutral $FeCl_3$
D.	Benzoic acid + Cinnamic acid	IV.	Ammonical silver nitrate

Choose the **correct** answer from the options given below :

Options :

691121235. A-IV, B-II, C-I, D-III

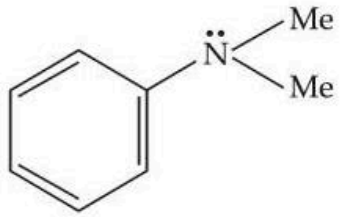
691121236. A-IV, B-II, C-III, D-I

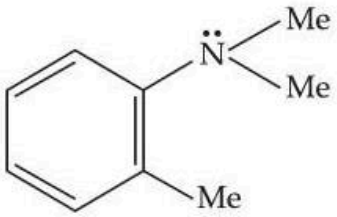
691121237. A-II, B-IV, C-I, D-III

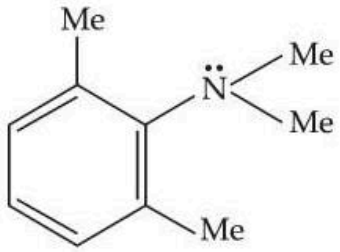
691121238. A-II, B-IV, C-III, D-I

Question Number : 68 Question Id : 69112168 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the three aromatic molecules (P, Q and R) whose structures have been given below :

P. 

Q. 

R. 

The **correct** order regarding the reactivity of these compounds with $Ph-\underset{(+)}{N}\equiv N\,Cl^{(-)}$ under optimum but slightly acidic medium is :

Options :

691121239. P > Q > R

691121240. R > P > Q

691121241. R > Q > P

691121242. $P > R > Q$

Question Number : 69 Question Id : 69112169 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

	List - I **Vitamin**		**List - II** **Name**
A.	Vitamin B_1	I.	Pyridoxine
B.	Vitamin B_2	II.	Ascorbic acid
C.	Vitamin B_6	III.	Thiamine
D.	Vitamin C	IV.	Riboflavin

Choose the **correct** answer from the options given below :

Options :

691121243. A-II, B-I, C-III, D-IV

691121244. A-IV, B-III, C-II, D-I

691121245. A-III, B-IV, C-I, D-II

691121246. A-I, B-III, C-II, D-IV

Question Number : 70 Question Id : 69112170 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A salt with few drops of conc. HCl gives apple green colour in flame test. The group precipitate of the salt is dissolved in acetic acid and treated with K_2CrO_4 to give yellow precipitate. When the sodium carbonate extract of the salt solution is heated with conc. HNO_3 and ammonium molybdate, it resulted a canary yellow precipitate. The cation and anion present in the salt are respectively,

Options :

691121247. Ca^{2+} and SO_4^{2-}

691121248. Ba^{2+} and PO_4^{3-}

691121249. Mn^{2+} and PO_4^{3-}

691121250. Ba^{2+} and SO_4^{2-}

Chemistry Section B

Section Id :	6911216
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory

Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	6911216
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Id : 69112171 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

5.33 g of $CrCl_3 \cdot 6H_2O$, which is a 1 : 3 electrolyte, is dissolved in water and is passed through a cation exchanger. The chloride ions in the eluted solution, on treatment with $AgNO_3$ results in 8.61 g of AgCl. The ratio of moles of complex reacted and moles of AgCl formed is _____ $\times 10^{-2}$. (Nearest integer)

[Molar mass in g mol^{-1} Cr : 52, Ag : 108, Cl : 35.5, H : 1, O : 16]

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 72 Question Id : 69112172 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the isomers of hydrocarbon with molecular formula C_5H_{10}. These isomers do not decolourise $KMnO_4$ solution. These isomers are subjected to chlorination with chlorine in presence of light to give monochloro compounds. The total number of monochloro compounds (structural isomers only) formed is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 73 Question Id : 69112173 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

One mole of an alkane (x) requires 8 mole oxygen for complete combustion. Sum of number of carbon and hydrogen atoms in the alkane (x) is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 74 Question Id : 69112174 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

For reaction A $\rightarrow$ P, rate constant $k = 1.5 \times 10^3\ s^{-1}$ at 27°C

If activation energy for the above reaction is 60 kJ mol^{-1}, then the temperature (in °C) at which rate constant, $k = 4.5 \times 10^3\ s^{-1}$ is _____. (Nearest integer)

Given : log 2 = 0.30, log 3 = 0.48, R = 8.3 J K^{-1} mol^{-1}, ln 10 = 2.3

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 75 Question Id : 69112175 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

At the transition temperature T, $A \rightleftharpoons B$ and $\Delta G^0 = 105 - 35 \log T$ where A and B are two states of substance X. The transition temperature in °C when pressure is 1 atm is _____. (Nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1